Reliability-aware active learning for FFLO phase spaces

Finite-momentum superconducting phase diagrams require repeated minimization over pairing amplitude and momentum, followed by spectral tests wherever topology is meaningful. We combine an exact Bogoliubov–de Gennes oracle with active learning that schedules, but never replaces, these calculations. The method reconstructs the two-dimensional Normal, uniform-superconducting, and FFLO regions with a sparse allocation of exact calculations. Restricting topology to reliable gapped FFLO states reveals connected conventional and topological FFLO support separated by a numerically reconstructed gap-closing region. Extending this hierarchy to $(k_BT/t, J_A/t, \mu/t)$ exposes systematic thermodynamic and topology-aware structure across a finite sampled domain. The latter evidence is reported as discrete support and narrow-μ bins, not as a converged continuous topological surface or complete three-dimensional atlas.

INTRODUCTION

Superconductivity with nonzero center-of-mass pairing momentum provides a route beyond a spatially uniform condensate. Fulde and Ferrell and, independently, Larkin and Ovchinnikov established the finite-momentum pairing and spatially modulated mechanisms now grouped under the FFLO setting [1, 2]. Subsequent reviews have developed the thermodynamic, material, and ultracold-gas consequences of Fermi-surface mismatch and competing pairing momenta [3–5]. The central phase-selection problem is therefore intrinsically thermodynamic: competing minima must be compared over both pairing amplitude and momentum.

The search for FFLO structure spans several physical platforms. Heavy-fermion and organic superconductors have displayed field-induced phases and spectroscopic, calorimetric, or magnetic signatures discussed in relation to FFLO order [6–10]; these representative studies motivate the problem without implying that every identification is universal or uncontested. Population-imbalanced ultracold Fermi gases provide a complementary tunable setting for pairing under mismatch, phase separation, and finite-momentum phases [11–15].

Altermagnets add a distinct route to this competition. Their compensated magnetic order can nevertheless generate symmetry-structured momentum-dependent spin splitting [16–19]. Model studies have connected this splitting to zero-field finite-momentum or other unconventional pairing states [20–22], and neighboring altermagnetic constructions have been proposed for topological superconductivity and Majorana modes [23, 24]. These precedents motivate an altermagnetic finite-q problem but do not determine the phase structure of the one-dimensional model studied here.

Topology adds a second, logically separate requirement. Finite-momentum pairing can coexist with topological superfluidity in spin-orbit-coupled models [25–27]. More generally, bulk symmetries and invariants organize topological superconducting phases [28–32]. Majorana-platform studies connect those bulk distinctions to boundary physics [33–36]. In a gapped one-dimensional Bogoliubov–de Gennes (BdG) system in symmetry class D, the relevant $\mathbb{Z}_2$ distinction can be expressed through a Pfaffian invariant [37, 38]; its use is meaningful only after bulk-gap applicability is established. A topology-resolved FFLO map must therefore identify a reliable thermodynamic minimum before classifying trusted gapped FFLO states.

This hierarchy is computationally demanding because every accepted parameter point requires minimization of $F(\Delta, q)$, comparison with the normal reference, adaptive checks of shallow or edge minima, and, where applicable, full-zone spectral and Pfaffian diagnostics. Active learning and Bayesian experimental design formalize the selection of informative sequential evaluations [39–41]; phase-diagram applications demonstrate boundary-focused sampling [42–44], while adaptive sampling has also reduced expensive electronic-structure, atomistic, and chemical search budgets [45–50]. For a physical phase diagram, however, an acquisition model should decide where to calculate next, not replace the exact calculation that defines the phase.

Here we combine an exact BdG oracle with active learning to resolve the thermodynamic and topology-aware structure of an altermagnetic finite-momentum superconductor. The oracle supplies every physical label, while the surrogate redirects subsequent evaluations toward uncertain or rapidly varying regions. This strategy reconstructs the two-dimensional thermodynamic phases from a sparse subset of the prescribed candidate locations, resolves connected cFFLO and tFFLO support within the reliable gapped FFLO sector, and exposes systematic structure across a finite three-dimensional domain. The three-dimensional evidence remains a discrete point cloud and narrow-μ bins rather than a converged continuous topological surface.

The detailed model, algorithms, counting conventions, and numerical diagnostics are provided in the Supplemental Material [51].

RESULTS

Model and physical hierarchy.— We use $\lambda_y/t = \lambda_z/t = 0.6$ and $U/t = 1.6$; the two-dimensional results fix $\mu/t = 0.55$. The single-particle sector is $h(k) = t\cos k\,\sigma_0 +$

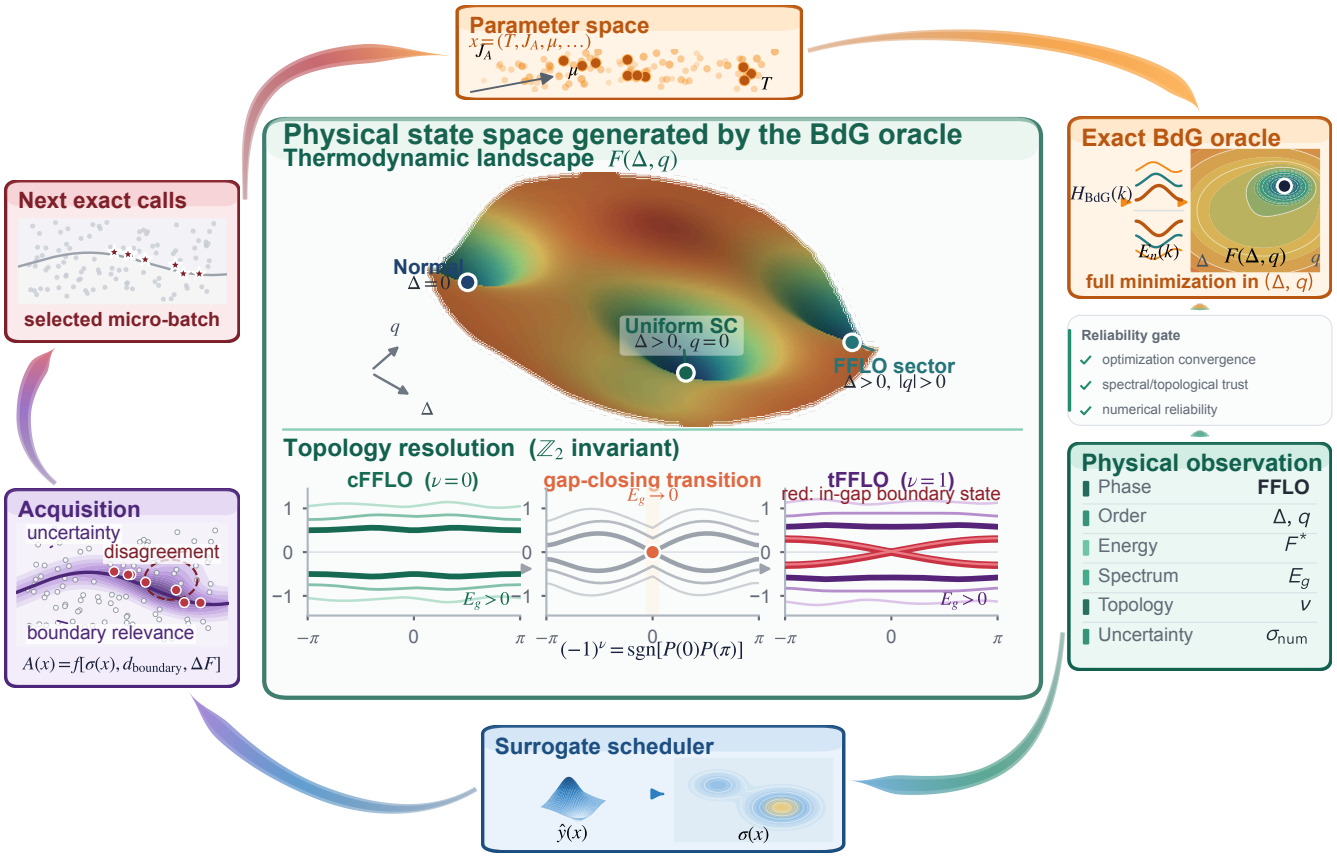


FIG. 1. **Physical hierarchy and adaptive scheduling.** Schematic illustration, not computed phase data: the altermagnetic and spin-orbit BdG model generates a normal-referenced $F_{\mathrm{impl}}(\Delta, q)$ landscape. Its reliable minimum determines Normal, uniform-SC, or FFLO order; only a gapped FFLO minimum proceeds to the Pfaffian test. The surrogate cycle proposes subsequent exact calls and does not determine either label.

$\lambda_y \sin k\, \sigma_y + [\lambda_z \sin k + J_A \cos k]\sigma_z$. The odd-in-$k$ spin-orbit fields and the altermagnetic $J_A \cos k$ splitting produce a momentum-dependent spin texture, while pairing connects states shifted by $\pm q/2$ in the Nambu Hamiltonian. A finite-momentum state therefore cannot be found by optimizing the order-parameter amplitude alone: for every external coordinate x the exact oracle determines

$$(\Delta_{\mathrm{opt}}, q_{\mathrm{opt}}) = \arg\min_{\Delta, q} F_{\mathrm{impl}}(\Delta, q), \tag{1}$$

where F_{impl} is the normal-referenced occupation-weighted spectral objective used by the implemented numerical calculation. Its reliable minimum gives Normal order when the optimum is unpaired, uniform superconductivity when $\Delta_{\mathrm{opt}} > 0$ and q_{opt} is zero within resolution, and FFLO order when both the amplitude and finite momentum are resolved.

Topology is a second, logically separate test. It is evaluated only for a thermodynamically accepted, numerically reliable FFLO minimum whose full Brillouin-zone spectrum is gapped. Under that applicability condition,

$$(-1)^\nu = \mathrm{sgn}[P(0)P(\pi)] \tag{2}$$

distinguishes cFFLO ($\nu = 0$) from tFFLO ($\nu = 1$); a change of index requires a bulk gap closing and reopening, not a change in basin depth or the sign of q. Figure 1 summarizes this hierarchy. Regression and classification ensembles see accepted exact observations and rank the next microbatch, but every physical label remains an output of the BdG objective and spectral tests.

Two-dimensional thermodynamic organization.— The exact observations in Fig. 2 organize into a curved outer envelope separating superconducting support below from Normal support above. At low temperature the superconducting sector reaches comparatively large J_A/t; as temperature rises, the largest sampled splitting that still supports pairing moves downward. Most of the paired region is finite-q FFLO. Uniform superconductivity instead forms a narrow strip close to $J_A = 0$, so the sampled state changes rapidly from zero- to finite-momentum pairing once the altermagnetic splitting is introduced. This thin

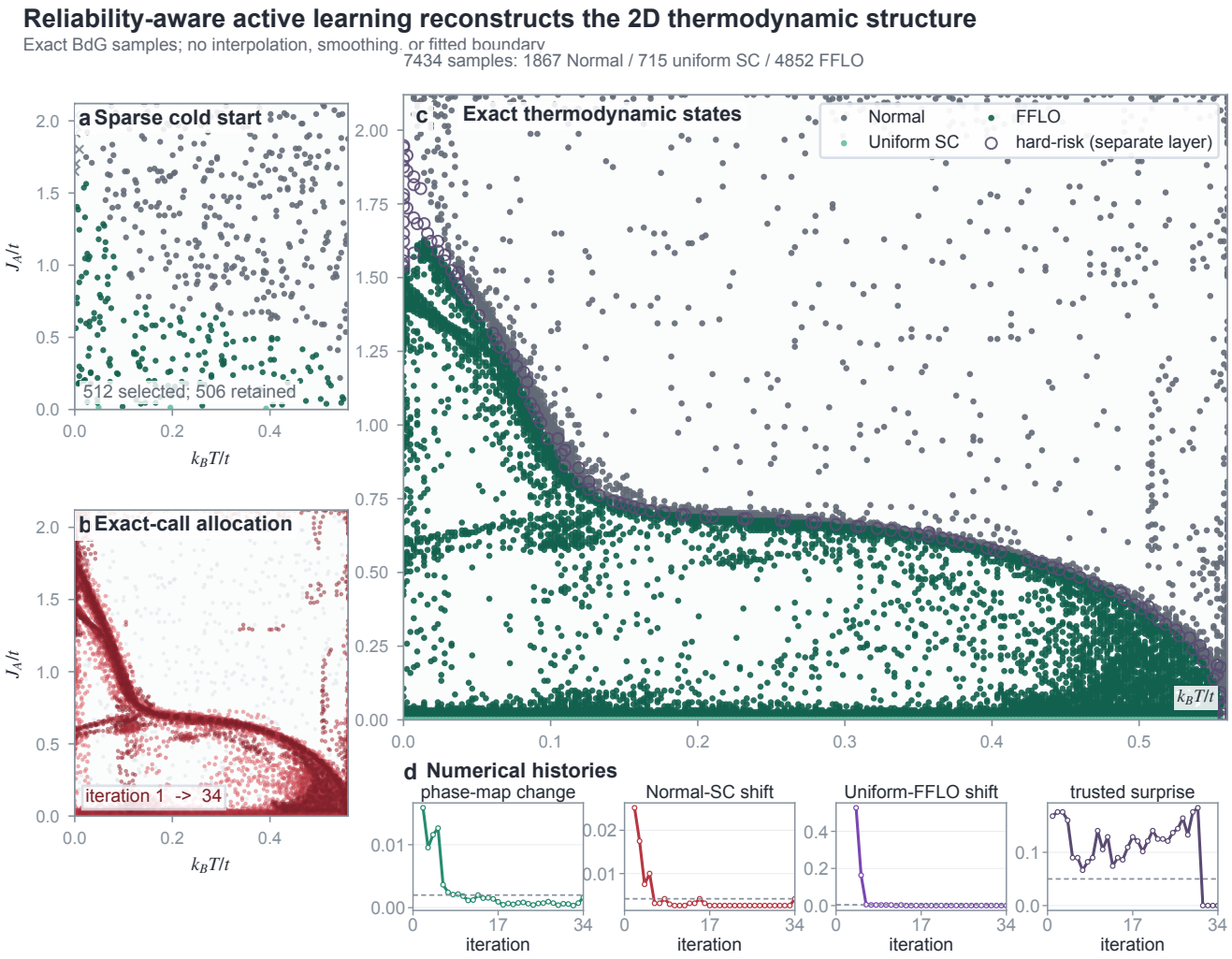


FIG. 2. **Two-dimensional thermodynamic reconstruction.** **(a)** Sparse exact initialization. **(b)** Exact-call allocation across Stage I and its Stage-II continuation; darker red denotes later iterations (1–34), not a phase or probability. **(c)** The 7,434-coordinate Stage-II endpoint, which contains the 6,880-coordinate Stage-I endpoint plus 554 additional exact coordinates; open symbols denote numerically flagged evaluations, not a phase. **(d)** Numerical histories; dashed lines are the saved operational references described in the Supplemental Material, not physical scales. The location-count comparison uses Stage I, not the Stage-II continuation.

inner structure and the descending outer envelope are precisely the regions where small changes of the optimized minimum can change a label, motivating concentrated exact minimization rather than a smooth surrogate boundary.

The search begins with a sparse random design over the full $(k_BT/t, J_A/t)$ domain. Subsequent calls accumulate along both thermodynamic structures while a continuing exploration channel samples their interiors and the remaining domain. In the primary Stage-I discovery, 6,880 unique exact coordinates are retained from 77,361 prescribed candidate locations, corresponding to an 8.89% location fraction and an 11.24-fold reduction relative to evaluating every grid location. The Stage-II reliability continuation adds exact evaluations without changing any shared phase label and supplies the 7,434-coordinate endpoint shown in panel (c). The efficiency ratio is therefore a Stage-I location count, not a wall-time acceleration or a comparison with matched random sampling. Selected requests and retained coordinates are distinguished in Supplemental Material, Table S5; neither is a complete execution ledger.

Topology-resolved FFLO structure.— Thermodynamic FFLO order alone does not specify the bulk topological sector. Applying the full-zone gap guard and Pfaffian invariant resolves the trusted FFLO observations into two gapped supports (Fig. 3). The cFFLO sector occupies lower J_A/t and extends to higher temperature, whereas the sampled tFFLO sector is shifted toward stronger altermagnetic splitting and lower temperature. Their

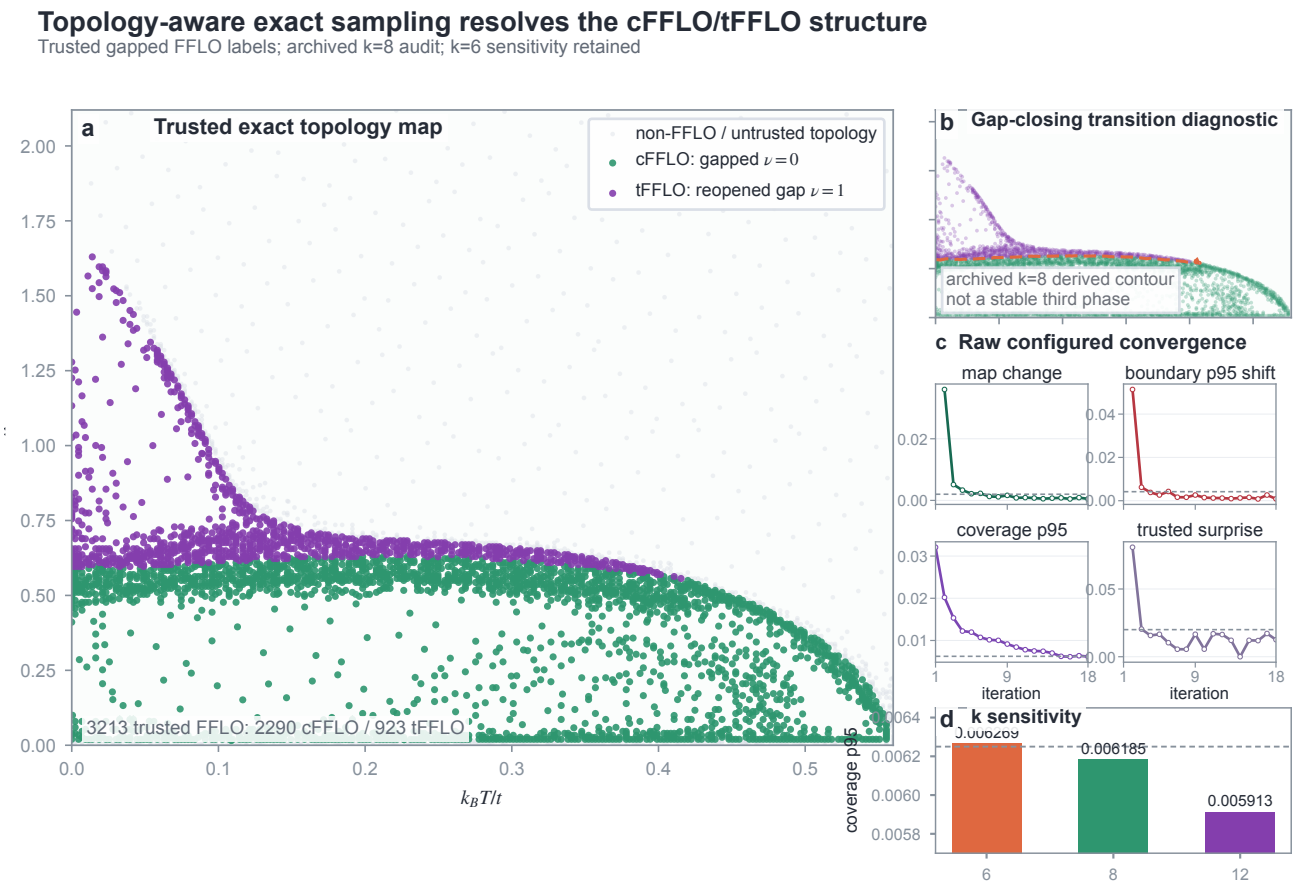


FIG. 3. **Bulk topology inside the thermodynamic FFLO sector. (a)** Trusted gapped FFLO observations resolved into cFFLO and tFFLO by the full-zone gap and Pfaffian invariant. **(b)** A support-restricted contour and local opposite-$\mathbb{Z}_2$ brackets locate a gap-closing region but not an exact boundary. **(c,d)** Numerical histories and neighborhood sensitivity (Supplemental Material, Sec. S9).

meeting region follows a curved locus where the bulk gap closes before reopening with the opposite Pfaffian product. Thus the topology-resolved map adds a bulk-state distinction that neither the position of the F_{impl} minimum nor the sign of q_{opt} can provide.

The exact labels are discrete, so the plotted transition line is a support-restricted reconstruction rather than an independently calculated phase boundary. Nearby observations with opposite $\mathbb{Z}_2$ indices bracket the transition region, and changing the reconstruction neighborhood produces a finite displacement of the guide. We therefore use the contour to organize the sampled close-and-reopen structure, not to claim an exact line or a stable third phase. The retained sensitivity also leaves open narrow nodal alternatives that the present sampling does not exhaust. This separation between exact bulk labels and a derived visual guide is carried into the higher-dimensional extension.

Chemical-potential-dependent higher-dimensional structure.— The four narrow-μ bins of exact observations in Fig. 4 show that the two-dimensional organization does not remain rigid as the third coordinate changes. At $\mu/t = -0.45$ and 0.15, sampled cFFLO support lies mainly at lower J_A/t, while tFFLO is concentrated toward higher splitting and lower temperature; in the latter cut the trusted tFFLO support is especially confined to low temperature. At $\mu/t = 0.85$, uniform-SC support extends over a broader J_A/t range and trusted tFFLO spans a wider part of the sampled slice. By $\mu/t = 1.45$, the observed paired support is restricted to low splitting and moderate-to-low temperature, and the narrow slice contains no trusted tFFLO observation. This last statement is about sampled support, not proof that tFFLO is absent elsewhere near that chemical potential.

The three-dimensional point cloud establishes the finite sampled domain and displays how exact thermodynamic and topology labels populate it; the narrow slices provide readable local cross-sections without imposing continuity between them. Selective evaluation is especially useful in this enlarged candidate space because it can retain direct BdG evaluation while directing calls toward changing structures. Nevertheless, nonuniform sampling

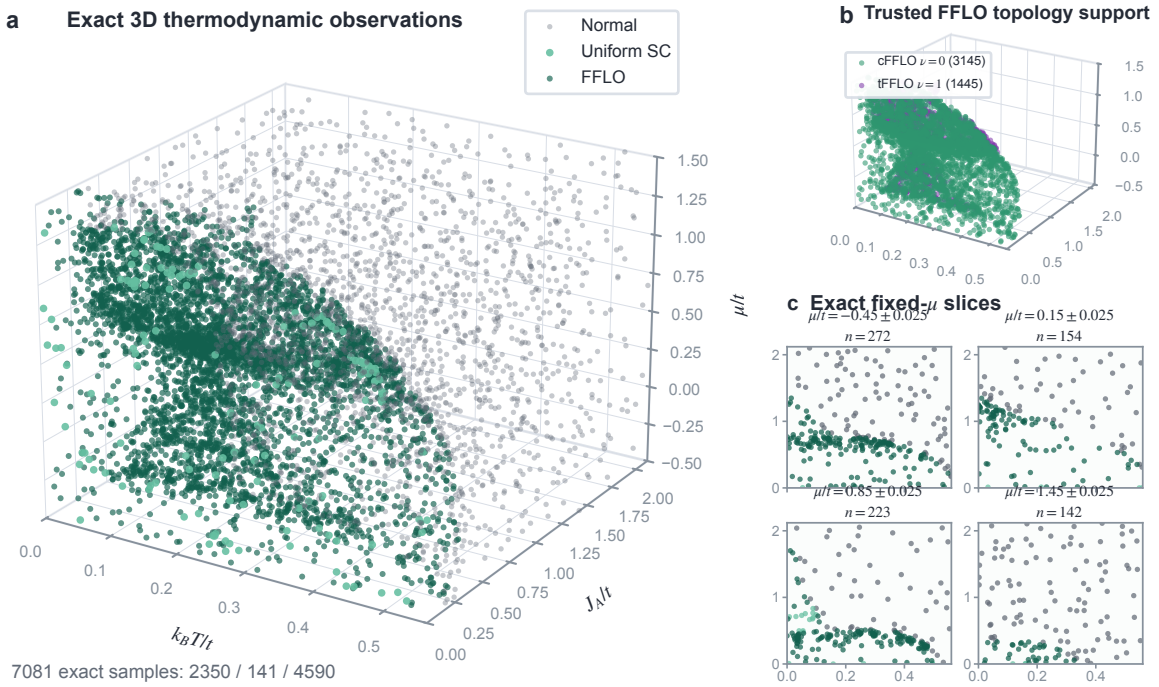


FIG. 4. **Finite-domain three-dimensional exploration.** **(a)** Exact thermodynamic observations in $(k_BT/t, J_A/t, \mu/t)$. **(b)** Discrete trusted FFLO topology support. **(c)** Four narrow-μ bins (centers $-0.45, 0.15, 0.85, 1.45$, half-width 0.025) without interpolation; see Supplemental Material, Sec. S10. Slice populations reflect adaptive sampling and are not phase-volume estimates; the four cuts do not define a continuous topological surface.

prevents phase counts from being interpreted as thermodynamic volumes, and four cuts cannot determine a continuous boundary or topology surface. Figure 4 therefore demonstrates scalable higher-dimensional exploration and chemical-potential-dependent reorganization, not a complete three-dimensional atlas or a fully resolved new mechanism.

DISCUSSION

The physical result is a hierarchy of structure rather than a single phase map. The two-dimensional exact data show a narrow uniform-SC strip, an extended finite-q region, and a curved Normal–superconducting envelope; the spectral analysis then resolves the FFLO sector into gapped conventional and topological bulk supports. Varying chemical potential reorganizes the coordinate support of those states and can remove trusted tFFLO observations from an individual sampled slice. Topology therefore supplies information that a thermodynamic FFLO label alone cannot encode.

Methodologically, reliability-aware active learning differs from a black-box phase classifier because physical labels are obtained from the BdG calculation at every layer. Numerical reliability is stored separately from phase identity, topology is masked outside reliable gapped FFLO states, and uncertainty affects where the next calculation is performed rather than what phase is declared. This organization makes the expensive minimization and spectral tests selectively deployable without turning the surrogate into a physics solver.

The location-count economy does not establish wall-time acceleration, because no matched exhaustive timing baseline was established, and it does not establish boundary enrichment, because boundary membership was not recorded for the complete candidate set. The stopping settings diagnose a particular numerical workflow rather than universal physical scales. Likewise, the three-dimensional evidence is discrete and nonuniformly sampled, so it supports higher-dimensional exploration but neither phase volumes nor a closed continuous atlas.

The method combines adaptive sampling with direct BdG evaluation while keeping numerical reliability separate from phase identity. The results establish finite thermodynamic and topology-resolved sampled supports; determining continuous boundaries requires additional evidence beyond those observations.

Data availability. The numerical data and source configurations are preserved in the project archive. No public repository or persistent identifier has been established for this review version.

[1] P. Fulde and R. A. Ferrell, Physical Review **135**, A550 (1964).
[2] A. I. Larkin and Y. N. Ovchinnikov, Soviet Physics JETP **20**, 762 (1965).
[3] R. Casalbuoni and G. Nardulli, Reviews of Modern Physics **76**, 263 (2004).
[4] Y. Matsuda and H. Shimahara, Journal of the Physical Society of Japan **76**, 051005 (2007).
[5] J. J. Kinnunen, J. E. Baarsma, J.-P. Martikainen, and P. Törmä, Reports on Progress in Physics **81**, 046401 (2018).
[6] H. A. Radovan, N. A. Fortune, T. P. Murphy, S. T. Hannahs, E. C. Palm, S. W. Tozer, and D. Hall, Nature **425**, 51 (2003).
[7] A. Bianchi, R. Movshovich, C. Capan, P. G. Pagliuso, and J. L. Sarrao, Physical Review Letters **91**, 187004 (2003).
[8] M. Kenzelmann, T. Strässle, C. Niedermayer, M. Sigrist, B. Padmanabhan, M. Zolliker, A. D. Bianchi, R. Movshovich, E. D. Bauer, J. L. Sarrao, and J. D. Thompson, Science **321**, 1652 (2008).
[9] R. Lortz, Y. Wang, A. Demuer, P. H. M. Böttger, B. Bergk, G. Zwicknagl, Y. Nakazawa, and J. Wosnitza, Physical Review Letters **99**, 187002 (2007).
[10] H. Mayaffre, S. Krämer, M. Horvatić, C. Berthier, K. Miyagawa, K. Kanoda, and V. F. Mitrović, Nature Physics **10**, 928 (2014).
[11] Y. an Liao, A. S. C. Rittner, T. Paprotta, W. Li, G. B. Partridge, R. G. Hulet, S. K. Baur, and E. J. Mueller, Nature **467**, 567 (2010).
[12] M. W. Zwierlein, A. Schirotzek, C. H. Schunck, and W. Ketterle, Science **311**, 492 (2006).
[13] G. B. Partridge, W. Li, R. I. Kamar, Y. an Liao, and R. G. Hulet, Science **311**, 503 (2006).
[14] D. E. Sheehy and L. Radzihovsky, Annals of Physics **322**, 1790 (2007).
[15] L. Radzihovsky and D. E. Sheehy, Reports on Progress in Physics **73**, 076501 (2010).
[16] L. Šmejkal, J. Sinova, and T. Jungwirth, Physical Review X **12**, 040501 (2022).
[17] L. Šmejkal, J. Sinova, and T. Jungwirth, Physical Review X **12**, 031042 (2022).
[18] I. Mazin, Physical Review X **12**, 040002 (2022).
[19] L. Šmejkal, R. González-Hernández, T. Jungwirth, and J. Sinova, Science Advances **6**, eaaz8809 (2020).
[20] D. Chakraborty and A. M. Black-Schaffer, Physical Review B **110**, L060508 (2024).
[21] S. Hong, M. J. Park, and K.-M. Kim, Physical Review B **111**, 054501 (2025).
[22] A. Bose, S. Vadnais, and A. Paramekanti, Physical Review B **110**, 205120 (2024).
[23] D. Zhu, Z.-Y. Zhuang, Z. Wu, and Z. Yan, Physical Review B **108**, 184505 (2023).
[24] S. A. A. Ghorashi, T. L. Hughes, and J. Cano, Physical Review Letters **133**, 106601 (2024).
[25] W. Zhang and W. Yi, Nature Communications **4**, 2711 (2013).
[26] C. Qu, Z. Zheng, M. Gong, Y. Xu, L. Mao, X. Zou, G. Guo, and C. Zhang, Nature Communications **4**, 2710 (2013).
[27] Y. Cao, S.-H. Zou, X.-J. Liu, S. Yi, G.-L. Long, and H. Hu, Physical Review Letters **113**, 115302 (2014).
[28] N. Read and D. Green, Physical Review B **61**, 10267 (2000).
[29] M. Sato and Y. Ando, Reports on Progress in Physics **80**, 076501 (2017).
[30] X.-L. Qi and S.-C. Zhang, Reviews of Modern Physics **83**, 1057 (2011).
[31] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig, Physical Review B **78**, 195125 (2008).
[32] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, New Journal of Physics **12**, 065010 (2010).
[33] R. M. Lutchyn, J. D. Sau, and S. D. Sarma, Physical Review Letters **105**, 077001 (2010).
[34] Y. Oreg, G. Refael, and F. von Oppen, Physical Review Letters **105**, 177002 (2010).
[35] J. Alicea, Reports on Progress in Physics **75**, 076501 (2012).
[36] C. Beenakker, Annual Review of Condensed Matter Physics **4**, 113 (2013).
[37] A. Y. Kitaev, Physics-Uspekhi **44**, 131 (2001).
[38] J. C. Budich and E. Ardonne, Physical Review B **88**, 075419 (2013).
[39] D. Cohn, L. Atlas, and R. Ladner, Machine Learning **15**, 201 (1994).
[40] D. J. C. MacKay, Neural Computation **4**, 590 (1992).
[41] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, Proceedings of the IEEE **104**, 148 (2016).
[42] K. Terayama, R. Tamura, Y. Nose, H. Hiramatsu, H. Hosono, Y. Okuno, and K. Tsuda, Physical Review Materials **3**, 033802 (2019).
[43] C. Dai, I. R. Bruss, and S. C. Glotzer, The Journal of Physical Chemistry B **124**, 1275 (2020).
[44] T. Lookman, P. V. Balachandran, D. Xue, and R. Yuan, npj Computational Materials **5**, 21 (2019).
[45] K. Tran and Z. W. Ulissi, Nature Catalysis **1**, 696 (2018).
[46] E. V. Podryabinkin and A. V. Shapeev, Computational Materials Science **140**, 171 (2017).
[47] R. Jinnouchi, F. Karsai, and G. Kresse, Physical Review B **100**, 014105 (2019).
[48] J. Vandermause, S. B. Torrisi, S. Batzner, Y. Xie, L. Sun, A. M. Kolpak, and B. Kozinsky, npj Computational Materials **6**, 20 (2020).
[49] R. Batra, L. Song, and R. Ramprasad, Nature Reviews Materials **6**, 655 (2021).
[50] F. Noé, S. Olsson, J. Köhler, and H. Wu, Science **365**, eaaw1147 (2019).
[51] See Supplemental Material at [URL will be inserted by publisher] for the model, stage-specific algorithms, numerical diagnostics, and selection accounting.